Petty Cash Replenishment and JV Manager

A web-based Information System for Petty Cash Replenishment and JV Creation

(System Documentation)

Submitted By:

Averion, Jesica P.

Evangelista, Geraldine B.

Ortiz, Maria Concepcion A.

Toleos, Cesilyn Q.

Yaneza, Jame Rachel A.

CMS BSAIS BLOC 1

Submitted to:

Mr. Hensonn Palomado

Instructor

I. System Description

The Petty Cash Replenishment and JV Manager System is a web-based information system designed to automate and improve the petty cash reimbursement process of BotiCARD Inc. The current process relies heavily on physical documents, manual validation, and courier services for transmitting reimbursement requests from branches located across Luzon, Visayas and Mindanao to the Head Office. These practices often result in delays, incomplete submissions, additional costs, and difficulty in tracking reimbursement requests.

The proposed system digitizes the entire reimbursement workflow by enabling Branch Staff to create reimbursement requests, encode expense details, upload supporting documents, and submit requests electronically. The Cluster Manager can review, validate, approve, void, and post requests online, while the Finance Department can process approved reimbursements, generate Journal Vouchers (JV), record deposits, and maintain accounting records within the system.

The system also provides real-time status tracking, document management, notifications, audit trails, reporting capabilities, and centralized record storage. By eliminating reliance on courier services and paper-based processing, the system enhances efficiency, accuracy, accountability, and internal controls while supporting faster reimbursement processing and improved financial record management.

II. System Objectives

To develop a web-based Petty Cash Replenishment and JV System that automates the petty cash reimbursement process, improves operational efficiency, strengthens internal controls, and provides effective monitoring of reimbursement transactions at BotiCARD Inc.

Specific Objectives

1. To digitize the submission of petty cash reimbursement requests and supporting documents and eliminate the dependence on physical document transmission.
2. To reduce reimbursement processing time by providing an electronic workflow for submission, validation, approval, posting, and finance processing.
3. To minimize errors and incomplete submissions through validation rules, mandatory fields, and document upload requirements.
4. To provide real-time status tracking that allows Branch Staff, Cluster Managers, and Finance personnel to monitor request progress from submission to completion.

5. To centralize reimbursement records, supporting documents, approvals, Journal Vouchers, and deposit information in a single database.
6. To improve communication between Branch Staff, Cluster Managers, and the Finance Department through notifications and status updates.
7. To establish a complete audit trail that records all actions, approvals, corrections, postings, and financial transactions for accountability and audit purposes.
8. To support the Finance Department in preparing Journal Vouchers and recording reimbursement deposits accurately and efficiently.
9. To generate reports and transaction records that support monitoring, decision-making, compliance, and financial management requirements.
10. To reduce administrative expenses, courier costs, and paperwork associated with the manual petty cash reimbursement process.

III. Business – workplace process

1. Specific process, purpose and users/stakeholders.

Process Name: Petty Cash Replenishment and Reimbursement. The petty cash replenishment process allows branches to recover the money they have spent on authorized operating expenses. Its purpose is to ensure that expenses are properly documented, reviewed, approved, and recorded in the accounting system before the reimbursement is released.

The main users and stakeholders involved in the process are:

Branch Staff – Records petty cash expenses, prepares the reimbursement summary, and submits the required supporting documents.

Courier Service – Transports physical documents from the branch to the Cluster Manager in the current manual process.

Cluster Manager – Checks the accuracy and completeness of the reimbursement request, approves valid submissions, and forwards them to Finance.

Finance Department – Conducts the final review, prepares the System Journal Voucher (JV), and processes the reimbursement. BotiCARD Management – Establishes policies and controls to ensure that company funds are properly managed.

2. Current workflow in chronological steps, including forms, approvals and handoffs.

The current process relies mainly on physical documents, manual checking, and the transfer of documents between the branch, Cluster Manager, and Finance Department.

The workflow follows these chronological steps:

1. Expenses are incurred. Branch staff use petty cash for authorized branch expenses and determine when replenishment is needed.

2. Running balance is checked. The staff verifies the remaining petty cash balance to determine the amount that needs to be replenished.

3. Supporting documents are gathered. The staff collects petty cash vouchers, official sales invoices, paid utility bills, and other required documents to support the expenses.

4. Reimbursement summary is prepared. The staff compiles the expenses and prepares a summary showing the amounts being requested for reimbursement.

5. Documents are sent through a courier. The reimbursement summary and supporting documents are packaged and sent to the Cluster Manager through a courier service, such as LBC or JRS Express.

6. Cluster Manager receives and validates the documents. Upon receiving the package, the Cluster Manager checks the expense summary, amounts, vouchers, invoices, and other supporting documents for accuracy, completeness, and compliance with company policies.

7. Corrections are requested when necessary. If the documents contain errors or missing information, the Cluster Manager informs the branch staff. The staff corrects the discrepancies, attaches the necessary documents, and resends the package for another review.

8. The reimbursement is approved. If the documents are complete and accurate, the Cluster Manager approves the request and forwards the validated documents to the Finance Department.

9. Finance conducts the final review. Finance receives the approved request and verifies the total reimbursement amount and supporting documents.

10. System Journal Voucher is prepared. Once the request passes the final review, Finance prepares and records the System Journal Voucher in the accounting system.

11. Reimbursement is released. Finance deposits the approved reimbursement amount into the branch& #39; s account, replenishing its petty cash fund and completing the process.

3. Actual problems in the current process, their effects and possible causes.

The following are potential problems in the current process based on the described workflow. Their frequency and actual impact should be confirmed through workplace observation or transaction records.

Problem	Effect on the Process	Possible Cause
1. Delays in document Deliver	Reimbursement requests longer to reach the Cluster Manager, which may delay validation and replenishment.	Physical documents must be packaged and transported through a courier service.
2. Incomplete or incorrect supporting documents	Requests are returned for correction, increasing	Manual preparation and checking may result in

	processing time and the workload of both branch, staff and the Cluster Manager.	missing receipts, incorrect amounts, or incomplete information.
3. Difficulty tracking request status	status Branch staff may need to follow up to determine whether a request has been received, reviewed, approved, or forwarded to Finance.	The process depends on physical handoffs and communication rather than a centralized tracking system.
4. Repeated manual review and processing	The Cluster Manager and Finance staff spend additional time checking documents, communicating corrections, and handling paperwork.	The process relies heavily on manual checking and physical document management.

4. Information system proposal.

We propose a Petty Cash Replenishment and Reimbursement Monitoring System integrated with, or developed as a module of, BotiCARD's accounting system. The system would allow branch staff to create reimbursement requests, upload supporting documents, and submit them electronically to the Cluster Manager and Finance Department.

Proposed System Feature	How It Addresses the Problem	Expected Benefit
Digital Reimbursement Request	Branch staff encode expense details and upload scanned vouchers, invoices, and paid utility bills instead of sending physical documents through a courier.	Faster submission and reduced courier costs and delays
Required Fields and Document Checklist.	The system checks whether required. Fewer incomplete requests and less rework. Information and supporting documents	Fewer incomplete requests and less rework.

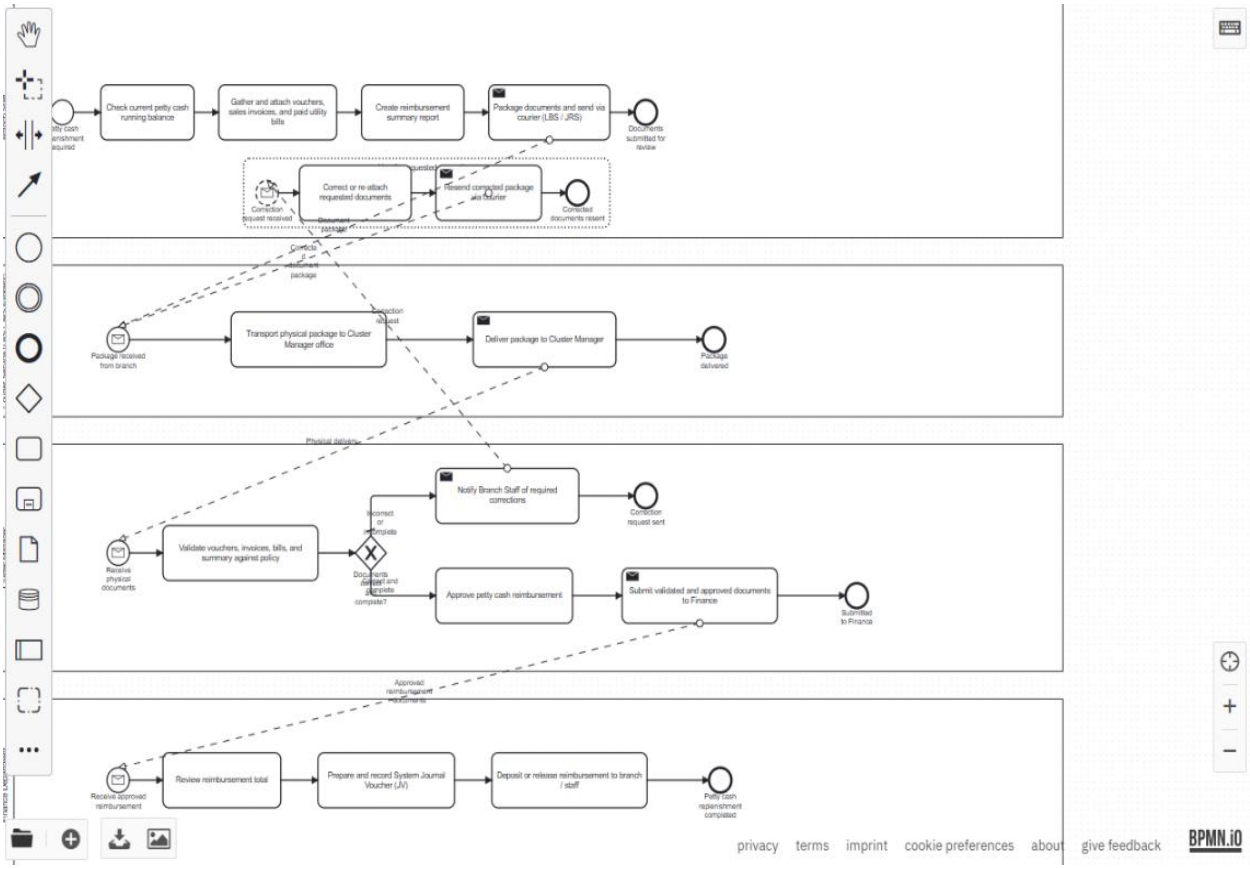
	are complete before submission.	
Electronic Review and Approval	The Cluster Manager reviews requests online and approves valid submissions or returns them with specific correction instructions.	Faster communication and a more organized review process.
Request Status Tracking	Each request receives a reference number and status, such as Submitted, Under Review, Returned for Correction, Approved or For Finance Processing.	Easier monitoring and fewer follow-up inquiries.
Centralized Digital Records	Requests, attachments, review actions, and approval history are stored in one system with appropriate access controls.	Easier document retrieval, improved traceability, and better audit readiness.
Finance Processing and Accounting Records	Finance accesses approved requests and records the corresponding System Journal Voucher in the accounting system.	

Problem-to-solution analysis

Current workplace problem	Operational effect	Proposed system improvement
1. Physical submission of reimbursement documents through courier services such as LBC or JRS Express.	Delays the delivery and review of requests and adds courier expenses.	Implement an electronic reimbursement request system where branch staff can submit requests and upload supporting documents online.
2. Incomplete or incorrect supporting documents, such as petty cash vouchers, sales invoices and paid utility bills.	Requests must be returned for correction, resulting in repeated work and longer processing time.	Add required fields and a document checklist to help ensure that requests are complete before submission.
3. Difficulty monitoring	Branch staff may need to follow up manually, while	Provide a tracking feature with a reference number

reimbursement status after documents are sent to the Cluster Manager or Finance Department.	the Cluster Manager may spend additional time checking request progress.	and status updates, such. as Submitted, Under Review, Returned for Correction, Approved, and Completed.
4. Risk of losing or misplacing physical documents.	Missing documents may delay reimbursement and make record retrieval difficult during audits.	Maintain centralized digital copies of reimbursement requests and supporting documents, with appropriate access controls and backup procedures.
5. Repeated manual checking and encoding of reimbursement details.	Increases the workload of the Cluster Manager and Finance staff and may result in encoding errors.	Use standardized electronic forms and, where feasible, integrate approved requests with the accounting system to reduce duplicate data entry.
6. Delays caused by repeated physical corrections and resubmissions.	Branch Staff must prepare and resend documents, extending the time needed to complete reimbursement.	Allow the Cluster Manager to return requests electronically with specific correction notes so branch staff can revise and resubmit them promptly.
7. Limited visibility of review and approval history.	It may be difficult to determine who reviewed a request, when it was approved, or where a delay occurred.	Maintain a digital audit trail recording submission dates, review actions, correction requests, approvals, and Finance processing updates.

IV. BPMN 2.0
BPMN Diagram (As-Is)



Step-by-Workflow Breakdown (Swimlane)

Process Name: Petty Cash Replenishment and Reimbursement

Participants (Pools / Swimlanes):

1. Branch Staff
2. Courier Service (LBS / JRS Express)
3. Cluster Manager
4. Finance Department

Swimlane 1: Branch Staff

1. Start Event: Petty cash replenishment is required.

2. Check Running Balance: Verify the accuracy of the current petty cash running balance.

3. Gather & Attach Documents:

* Collect Petty Cash Vouchers.

* Collect Sales Invoices.

* Collect paid utility bills.

4. Create Reimbursement Summary: Compile the expenses into a summary report.

5. Send via Courier: Package the documents and send them to the Cluster Manager using a courier service (e.g., LBS or JRS Express).

Swimlane 2: Courier Service

6. Transit: Transport the physical package from the branch to the Cluster Manager' s office.

Swimlane 3: Cluster Manager

7. Receive Documents: Receive the physical package delivered by the courier.

8. Validate Details & Accuracy: Inspect the vouchers, invoices, utility bills, and summary against company policies.

9. Decision Gateway (Validation Check):

[If incorrect or incomplete]: Notify the Branch Staff of the required corrections. (Loop back to Branch Staff to re-attach/correct documents and resend).

[If correct and complete]: Approve the petty cash reimbursement.

10. Submit to Finance: Forward the validated and approved documents to the Finance Department.

Swimlane 4: Finance Department

11. Receive Validated Reimbursement: Accept the approved documents from the Cluster Manager.

12. Review Reimbursement Total: Perform a final review of the total reimbursement amount requested.

13. Prepare System Journal Voucher (JV): Encode and record the transaction in the accounting system via a Journal Voucher.

14.Process Disbursement: Deposit or release the reimbursement amount to the branch/staff.

15. End Event: Petty cash replenishment process is completed.

Purpose: I want to expedite the process by creating a Petty Cash Replenishment and JV System. The Petty cash reimbursement process will be systematized and will save cost in transporting hard copy of supporting documents for the replenishment.

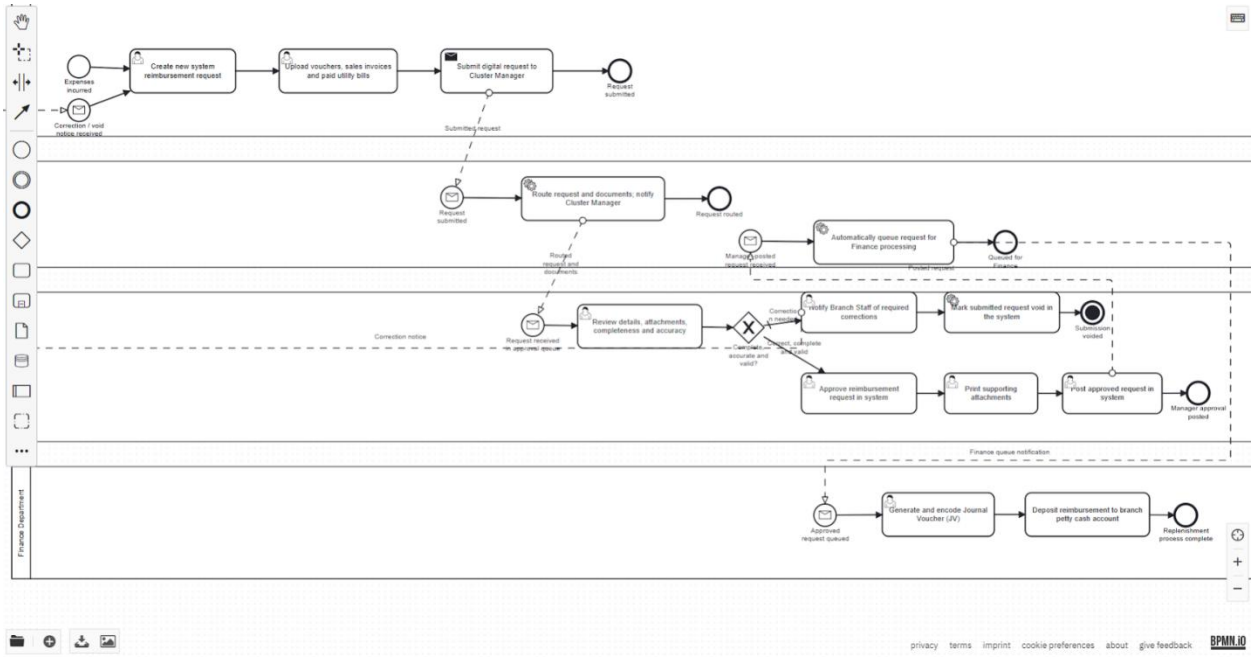
User roles:

Branch Staff: create, edit (while Draft), upload documents, submit, track own requests.

Cluster Manager: view requests of branches in the cluster, validate, approve, void, post.

Finance Department: view posted requests, create and post JVs, record deposits, view all records.

BPMN Diagram (To-Be)



Step-by-Workflow Breakdown (Swimlane)

Process Name:Systematized Petty Cash Replenishment and Reimbursement

Participants (Pools / Swimlanes):

1. Branch Staff
2. Accounting System (Automated/System Operations)
3. Cluster Manager
4. Finance Department

Swimlane 1: Branch Staff

1. **Start Event:** Expenses incurred by Branch Staff.
2. **Create System Reimbursement Request:** Encode the reimbursement request directly into the Accounting System.
3. **Upload Supporting Documents:** Scan and upload all required supporting documents (vouchers, sales invoices, paid utility bills) into the system request.
4. **Submit Request:** Transmit the digital reimbursement request to the Cluster Manager for approval.

Swimlane 2: Accounting System (Automated Transmittal)

5. **Transmit Request:** System routes the submitted request and supporting documents to the Cluster Manager's approval queue and notifies them.

Swimlane 3: Cluster Manager

6. **Receive Transmitted Request:** Access the pending reimbursement request via the Accounting System.
7. **Validate Reimbursement Request:** Review the digital details, attached documents, and figures for completeness and accuracy.
8. **Decision Gateway (Validation Check):**
    - **[If Correction Needed]:**
        - **Notify the Branch Staff of the required corrections**

        - **Void Request:** Mark the request as void in the system (terminating the request). (Loop back to Branch Staff to Create New System Reimbursement Request repeating the process)
        - **End Event (Voided):** Process terminates for the voided submission.
    - **[If Correct, Complete & Valid]:**
        - **Approve Reimbursement:** Systematically approve the request.
9. **Print Attachments:** Print physical copies of the required supporting attachments.
10. **Post Request:** Post the approved request in the system to finalize the manager-level approval step.

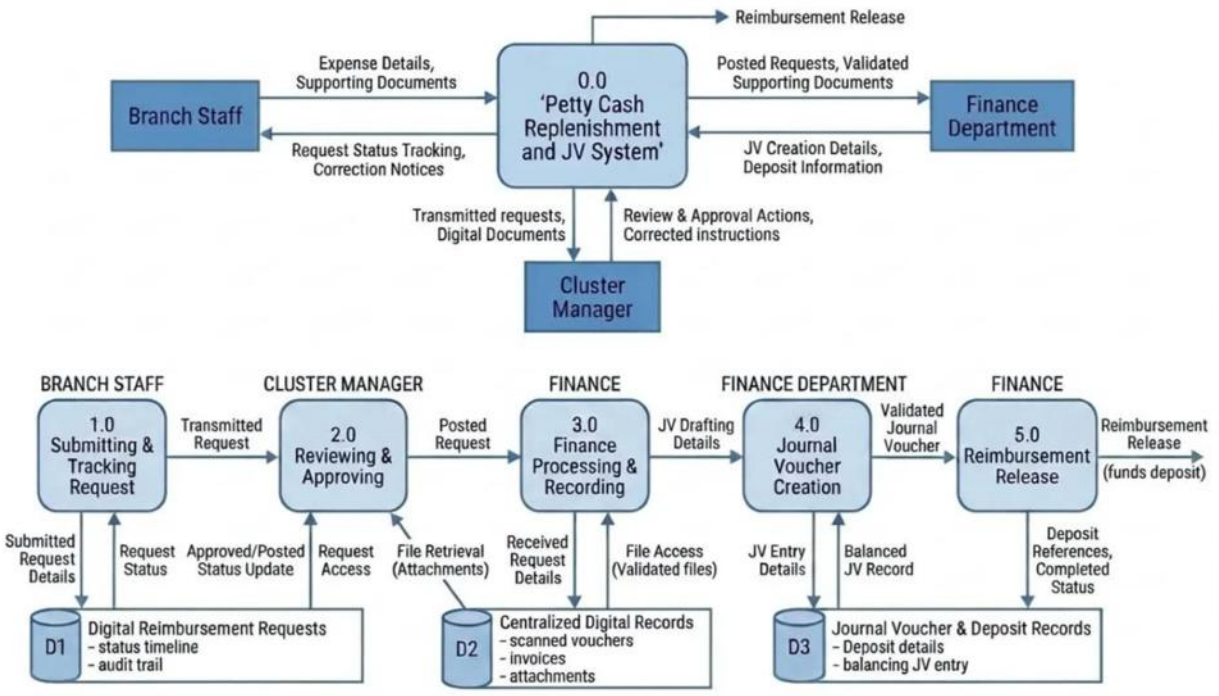
Swimlane 4: Finance Department

11. **Receive Approved Request:** System automatically queues the posted request for Finance processing.
12. **Create Journal Voucher (JV):** Systematically generate/encode the Journal Voucher for the petty cash reimbursement.
13. **Deposit Reimbursement Amount:** Transfer/deposit the approved reimbursement funds into the branch's petty cash account.
14. **End Event:** Systematized replenishment process complete.

Core workflow:

The systematized Petty Cash Replenishment Process begins when the Branch Staff incurs expenses and creates a System Reimbursement Request directly within the accounting system. She then uploads all complete supporting documents into the system and submits the digital request to the Cluster Manager. Upon receiving the transmitted request via the accounting system, the Cluster Manager proceeds to validate the details, completeness, and validity of the reimbursement request. If the request requires correction, the Cluster Manager voids the request in the system, which terminates that particular cycle. However, if the request is correct, complete, and valid, the Cluster Manager approves the reimbursement and posts the request in the system. Finally, the Finance Department receives the approved request, creates a Journal Voucher for the petty cash replenishment, and deposits the reimbursement amount to complete the process.

V. Data Flow Diagram



Level 0 - DFD

1. Context Diagram / Level-0 DFD

External Entities

1. Branch Staff

2. Cluster Manager

3. Finance Department

System Boundary

0.0 Petty Cash Replenishment and Reimbursement Monitoring System

Data Flows

Inputs to System

From Branch Staff:

o Expense Details (expense date, category, payee, ref no., amount)

o Supporting Documents (vouchers, sales invoices, paid utility bills)

From Cluster Manager:

o Validation Check Results (Approval or Void decision)

o Correction Details / Void Reason

From Finance Department:

o Journal Voucher Details (account codes, particulars, debit/credit values)

o Deposit Details (deposit date, bank reference, amount)

Outputs from System

To Branch Staff:

o System Notifications / Void Notice with Reason

o Request Tracking Status

To Cluster Manager:

o Transmitted Reimbursement Requests & Documents (in Cluster Queue)

To Finance Department:

o Posted Reimbursement Requests (in Finance Queue)

To Branch Account (External Fund Target):

Updated Petty Cash Balance / Reimbursement Release Confirmation

2. Level-0 DFD (Decomposed System Processes)

The system is broken down into three major logical processes as specified:

1. Process 1.0: Submit Request (Handled by Branch Staff & System Routing)

2. Process 2.0: Review Request (Handled by Cluster Manager)

3. Process 3.0: Track Request Status / Finance Processing (Handled by Finance Department & System Status Updates)

Process Details & Data Flows

Process 1.0: Submit Request

Primary Actor: Branch Staff

Description: Branch Staff creates a draft request, inputs expense lines, uploads

required supporting documents and submits the digital reimbursement request. The system transmits it to the Cluster Manager' s queue.

Inputs:

o Expense Details & Documents from Branch Staff

Outputs:

o Submitted/Transmitted Request to Data Store D1 (REIMB_REQUEST)

o Uploaded Attachments to Data Store D2 (SUPPORTING_DOCUMENT)

o Action Log Entry to Data Store D4 (APPROVAL_LOG)

o Pending Request Transmittal to Process 2.0

Process 2.0: Review Request

Primary Actor: Cluster Manager

Description: The Cluster Manager accesses transmitted requests, validates expense details against documents, and either approves, prints attachments, and posts the request, or issues a correction notice and voids the request.

Inputs:

o Pending Request Transmittal from Process 1.0

o Request Header & Lines from Data Store D1 (REIMB_REQUEST)

o Supporting Documents from Data Store D2 (SUPPORTING_DOCUMENT)

o Review Actions (Approve/Post/Void) from Cluster Manager

Outputs:

o Correction Notice & Void Reason to Data Store D3 (CORRECTION_NOTICE) and Branch Staff

o Approved/Posted Request Status to Data Store D1 (REIMB_REQUEST)

o Attachment Print Log to Data Store D4 (APPROVAL_LOG)

o Posted Request Notification to Process 3.0

Process 3.0: Track Request Status

Primary Actor: Finance Department / System Automation

Description: Finance receives posted requests, generates and balances the System Journal Voucher (JV), records deposit details (bank reference & deposit date), updates fund balances, and updates request timelines for user tracking.

Inputs:

o Posted Request Notification from Process 2.0

o Posted Request Data from Data Store D1 (REIMB_REQUEST)

o Account Mappings from Data Store D6 (EXPENSE_CATEGORY / CHART_OF_ACCOUNT)

o JV & Deposit Details from Finance Department

Outputs:

o Journal Voucher & JV Lines to Data Store D5 (JOURNAL_VOUCHER)

o Deposit Record & Balance Update to Data Store D7 (PETTY_CASH_FUND / DEPOSIT)

o Updated Status & Timeline Logs to Branch Staff, Cluster Manager, and Data Store D4 (APPROVAL_LOG)

3. Data Stores List (Level-0 DFD Data Stores)

D1: REIMB_REQUEST / REQUEST_LINE (Stores request headers, amounts, status, and expense lines)

D2: SUPPORTING_DOCUMENT (Stores uploaded vouchers, invoices, and utility bills)

D3: CORRECTION_NOTICE (Stores validation correction details for voided requests)

D4: APPROVAL_LOG / NOTIFICATION (Stores audit logs, timestamps, actions, and user notices)

D5: JOURNAL_VOUCHER / JV_LINE (Stores generated debit/credit accounting entries)

D6: EXPENSE_CATEGORY / CHART_OF_ACCOUNT (Stores account mapping data for JV creation).

D7: PETTY_CASH_FUND / DEPOSIT (Stores fund limits, running balances, and deposit transactions).

4. Input / Output Balance Check Matrix

External Entity	Input to System	Level-0 Process	Output from System	Level-0 Process
Branch Staff	Expense Lines & Documents	1.0 Submit Request	Notifications, Tracking Status & Void Notices	2.0 Review / 3.0 Track Status
Cluster Manager	Validation Check (Approve / Void / Post)	2.0 Review Request	Transmitted Request Queue & Documents	1.0 Submit Request
Finance Department	Journal Voucher Entries & Deposit			

Information flow: (DFD inputs and outputs)

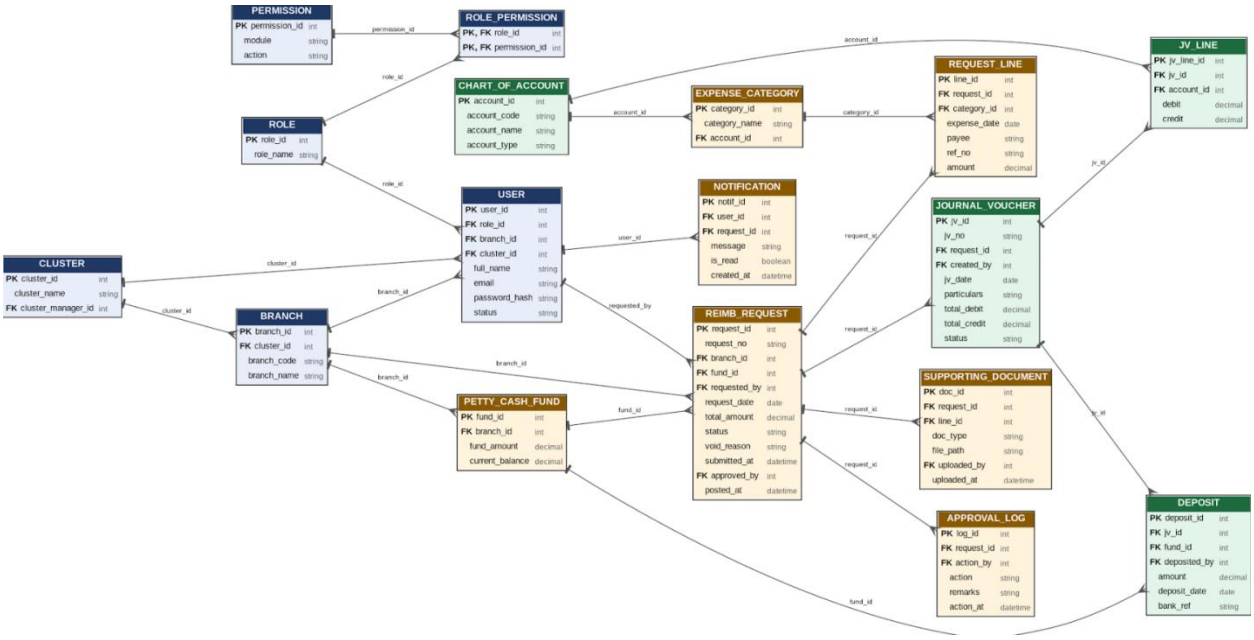
Process	Input	Output
1. Create request (Branch Staff)	Expense details, supporting documents	Draft request with lines and attachments
2. Submit and transmit (Accounting System)	Submitted request	Request in the Cluster Manager queue, notification, log entry
3. Validate (Cluster Manager)	Request, documents, figures	Approval, or a correction notice to Branch Staff followed by a void
4. Print and post (Cluster Manager)	Approved request	Printed attachments, posted request
5. Queue for Finance (Accounting System)	Posted request	Finance queue entry, log entry

6. Create JV (Finance)	Posted request, expense category to account mapping	Balanced Journal Voucher
7. Deposit (Finance)	JV, bank reference, deposit date	Deposit record, completed request

Data stores: requests (with lines, documents, JVs) and an activity log of every action. A voided request loops back to process 1 as a new request that points to the voided one.

Validation: Required to submit
• Request date.
• At least one expense line, each with date, category, payee, reference number and an amount above zero.
• At least one attached supporting document (voucher, sales invoice, paid utility bill).
• Total within the fund limit (P20,000 in the prototype).
Approval rules
• The Cluster Manager must tick all three checks (details correct, documents complete, amounts match receipts)
before approving.
• Voiding requires written corrections; the Branch Staff is notified.
• Attachments must be printed before posting.
• The Journal Voucher must balance (total debits equal total credits).
• Deposit requires a deposit date and a bank reference.
Status transitions (each action is allowed only for its role)

VI. Entity Relationship Diagram



Entities and fields

PK = primary key, FK = foreign key. Data types are proposed.

Users and access

ROLE - Roles: Branch Staff, Accounting System, Cluster Manager, Finance Department

Field	Key	Type	Notes
role id	PK	int	
Role_name	string	Branch Staff, Accounting System, Cluster Manager, Finance Department	

PERMISSION - Module and action permissions

Field	Key	Type	Notes
Permission _id	PK	int	
module	string		

action	string	Create, submit, transmit Approve, void, print, post, create, Jv deposit, view	

ROLE_PERMISSION - Links roles to permissions (many-to-many)

Field	Key	Type	Notes
role_id	PK FK	int	References ROLE
permission_id	PK, FK	int	References PERMISSION

CLUSTER - Group of branches under one Cluster Manager

Field	Key	Type	Notes
cluster_id	PK	int	
cluster_name	string		
cluster_manager_id	FK	int	References USER

BRANCH - Branch served by a cluster

Field	Key	Type	Notes
branch_id	PK	int	

cluster_id	FK	int	References CLUSTER
branch_code	string		
branch_name	string		

Petty Cash Replenishment and JV System - Data Model Part 2 -

USER - All users, including the automated Accounting System account

Field	Key	Type	Notes
User_id	PK	int	
role_id	FK	int	
branch_id	FK	int	
cluster_id	FK	int	
full_name	string		
email	string		
user_type	string	Human or System (Accounting System account)	
status	string	Active or inactive	

Request and transmittal

PETTY_CASH_FUND - One petty cash fund per branch

Field	Key	Type	Notes
fund_id	PK	int	
branch_id	FK	int	References BRANCH

			(unique)
fund_amount	decimal	Authorized fund	
current_balance	decimal	Updated on deposit	

Petty Cash Replenishment and JV System - Data Model Part 2

REIMB_REQUEST - Reimbursement request header; carries status and the void/replace link

Field	Key	Type	Notes
request_id	PK	int	
request_no	string	Unique, auto-generated	
previous_request_id	FK	int	Self-reference: the voided request this new request replaces
branch_id	FK	int	int References BRANCH
fund_id	FK	int	References PETTY_CASH_FUND
requested_by	FK	int	References USER (Branch Staff)
request_date	date		
total_amount	decimal	Sum of line amounts	
status	string	Draft, Submitted, Transmitted, Voided, Approved, Posted, JV	

		Created, Deposited	
submitted_at	datetime		
approved_by	FK	int	References USER (Cluster Manager)
approved_at	datetime		
posted_by	FK	int	References USER (Cluster Manager)
posted_by	FK	int	References USER (Cluster Manager)
posted_at	datetime		
voided_by	FK	int	References USER (Cluster
Void_at	datetime		
void_reason	string	Required when voided	

REQUEST_LINE - Expense lines of a request

Field	Key	Type	Notes
line_id	FK	int	
request_id	FK	int	References REIMB_REQUEST
category_id	FK	int	References EXPENSE_CATEGORY

expense_date	date		
payee	string		
ref_no	string		OR, SI, or voucher number
amount	decimal		

EXPENSE_CATEGORY - Expense categories mapped to accounts

Field	Key	Type	Notes
category_id	PK	int	
category_name	string		
account_id	FK	int	References CHART_OF_ACCOUNT; drives JV debit lines

Petty Cash Replenishment and JV System - Data Model Part 2

SUPPORTING_DOCUMENT - Uploaded vouchers, sales invoices, paid utility bills

Field	Key	Type	Notes
doc_id	PK	int	
request_id	FK	int	References REIMB_REQUEST
line_id	FK	int	References REQUEST_LINE (optional)
doc_type	string	Voucher, sales invoice, paid utility bill	

file_path	string		
uploaded_by	FK	int	References USER
uploaded_at	datetime		

TRANSMITTAL - System routing of a request to the Cluster Manager or Finance queue

Field	Key	Type	Notes
transmittal_id	PK	int	
request_id	FK	int	References REIMB_REQUEST
transmitted_by	FK	int	References USER (Accounting System account)
to_user_id	FK	int	References USER (Cluster Manager or Finance recipient)
stage	string	To Cluster Manager (step 5) or To Finance (step 11)	
queue_status	string	Pending, received, done	
transmitted_at	datetime		
received_at	datetime		

CORRECTION_NOTICE - Notice to Branch Staff of required corrections (before voiding)

Field	Key	Type	Notes

notice_id	PK	int	
request_id	FK	int	References REIMB_REQUEST (the request to be voided)
issued_by	FK	int	References USER (Cluster Manager)
issued_to	FK	int	References USER (Branch Staff
correction_details	string		
issued_at	datetime		
acknowledged_at	datetime		

Petty Cash Replenishment and JV System - Data Model Part 2

ATTACHMENT_PRINT_LOG - Record of printed supporting attachments by the Cluster Manager

Field	Key	Type	Notes
print_id	PK	int	
request_id	FK	int	References REIMB_REQUEST
doc_id	FK	int	References SUPPORTING_DOCUMENT
printed_by	FK	int	References USER (Cluster Manager
copies	int		
printed_at	datetime		

APPROVAL_LOG - Status history and audit trail of every action

Field	Key	Type	Notes
log_id	PK	int	
request_id	FK	int	References REIMB_REQUEST
action_by	FK	int	References USER (human or system)
action	string	Created, uploaded, submitted, transmitted, notified, voided, approved, printed, posted, queued, JV created, deposited	
remarks	string		
Action_at	datetime		

NOTIFICATION - System notifications to users

Field	Key	Type	Notes
notif_id	PK	int	
user_id	FK	int	References USER (recipient)
request_id	FK	int	References REIMB_REQUEST

notice_id	FK	int	
message	string		
is_read	boolean		
created_at	datetime		

Finance

CHART_OF_ACCOUNT - Accounts used in journal vouchers

Field	Key	Type	Notes
account_id	PK	int	
account_code	string		
account_name	string		
account_type	string		

Petty Cash Replenishment and JV System - Data Model Part 2

JOURNAL_VOUCHER - JV header for a replenishment

Field	Key	Type	Notes
jv_id	PK	int	
jv_no	string	Unique sequence	
request_id	FK	int	References REIMB_REQUEST (unique)
created_by	FK	int	References USER (Finance
jv_date	date		
particulars	string		
total_debit	decimal		

total_credit	decimal		
status	string	Draft or posted	

JV_LINE - Debit and credit lines of a JV

Field	Key	Type	Notes
jv_line_id	PK	int	
jv_id	FK	int	References JOURNAL_VOUCHER
account_id	FK	int	References CHART_OF_ACCOUNT
debit	decimal		
credit	decimal		

DEPOSIT - Reimbursement deposit to the branch petty cash account

Field	Key	Type	Notes
deposit_id	PK	int	
jv_id	FK	int	References JOURNAL_VOUCHER (unique)
fund_id	FK	int	References PETTY_CASH_FUND
deposited_by	FK	int	References USER (Finance)
amount	decimal		
deposit_date	date		
bank_ref	string		

Petty Cash Replenishment and JV System - Data Model Part 2

Primary key / foreign key relationships

Parent (PK side	Child (FK side	Cardinality	Foreign key
ROLE	USER	1 : many	USER.role_id
ROLE	ROLE_PERMISSION	1 : many	ROLE_PERMISSION.role_id
PERMISSION	ROLE_PERMISSION	1 : many	ROLE_PERMISSION.permission_id
CLUSTER	BRANCH	1 : many	BRANCH.cluster_id
CLUSTER	USER	1 : many	USER.cluster_id
BRANCH	USER	1 : many	USER.branch_id
BRANCH	PETTY_CASH_FUND	1 : 0..1	PETTY_CASH_FUND.branch_id
BRANCH	REIMB_REQUEST	1 : many	REIMB_REQUEST.branch_id
PETTY_CASH_FUND	REIMB_REQUEST	1 : many	REIMB_REQUEST.fund_id
USER	REIMB_REQUEST	1 : many	REIMB_REQUEST.requested_by
REIMB_REQUEST	REIMB_REQUEST	1 : 0..1	1 REIMB_REQUEST.previous_request_id
REIMB_REQUEST	REQUEST_LIN	1 : 1..many	REQUEST_LINE.request_id
EXPENSE_CATEGORY	REQUEST_LINE	1 : many	REQUEST_LINE.category_id
REIMB_REQUEST	SUPPORTING_DOCUMENT	1 : 1..many	SUPPORTING_DOCUMENT.request_id

REIMB_REQUEST	TRANSMITTAL	1 : 1..2	TRANSMITTAL.request_id
USER	TRANSMITTAL	1 : many	TRANSMITTAL.transmitted_by / to_user_id
REIMB_REQUEST	CORRECTION_NOTICE	1 : 0..many	CORRECTION_NOTICE.request_id
REIMB_REQUET	ATTACHMENT_PRINT_LOG	1 : 0..many	ATTACHMENT_PRINT_LOG.request_id
SUPPORTING_DOCUMENT	ATTACHMENT_PRINT_LOG	1 : 0..many	ATTACHMENT_PRINT_LOG.doc_id
REIMB_REQUEST	APPROVAL_LOG	1 : many	APPROVAL_LOG.request_id
USER	APPROVAL_LOG	1 : many	APPROVAL_LOG.action_by
USER	NOTIFICATION	1 : many	NOTIFICATION.user_id
CORRECTION_NOTICE	NOTIFICATION	1 : 0..many	NOTIFICATION.notice_id
CHART_OF_ACCOUNT	EXPENSE_CATEGORY	1 : many	EXPENSE_CATEGORY.account_id
CHART_OF_ACCOUNT	EXPENSE_CATEGORY	1 : many	EXPENSE_CATEGORY.account_id
REIMB_REQUEST	JOURNAL_VOUCHER	1 : 0..1	JOURNAL_OUCHER.request_id
JOURNAL_VOUCHER	JV_LINE	1 : 1..many	JV_LINE.jv_id
CHART_OF_ACCOU	JV_LINE	1 : many	JV_LINE.account_id

NT			
JOURNAL_VOUCHER	DEPOSIT	1 : 0..1	DEPOSIT.jv_id
PETTY_CASH_FUND	DEPOSIT	1 : many	DEPOSIT.fund_id

Role-based access (by user)

Entity	Branch Staff	Accounting System	Cluster Manager	Finance Department
REIMB_REQUEST	Create, edit while Draft, submit; own branch	Update status on transmit and queue	View cluster; approve, void, post	View posted onward; all branches

Petty Cash Replenishment and JV System - Data Model Part 2

Entity	Branch Staff	Accounting System	Cluster Manager	Finance Department
REQUEST_LINE, SUPPORTING_DOCUMENT	Add and edit while Draft	None	Read only	Read only
TRANSMITTAL	Read own	Create and update	Read; mark received	
TRANSMITTAL	Read own	Create and	Read; mark	Read; mark

		update	received	received
CORRECTION_NOTICE	Read own	Notify on creation	Create	Read
ATTACHMENT_PRINT_LOG	None	None	Read only	Create, edit before posting, post
APPROVAL_LOG, NOTIFICATION	Read own	Write all entries	Read cluster	Read all
PETTY_CASH_FUND	Read own branch	Update balance on deposit	Read cluster	Read all
USER, ROLE, PERMISSION	None	None	None	None (system admin only)

Integrity rules

Voided is final; a correction needs a new request linked by previous_request_id. A correction notice must exist before a request is voided, and a void requires a reason.A request needs at least one line and one supporting document before submission. A JV can be posted only when total debit equals total credit and equals the request total.

Unique: request_no, jv_no, JOURNAL_VOUCHER.request_id, DEPOSIT.jv_id, PETTY_CASH_FUND.branch_id.

Records are never deleted; changes are logged in APPROVAL_LOG. Row visibility is filtered by branch_id for staff and cluster_id for managers.

VII. System Features and Functionalities

#	Feature Area	Feature	Functionality Description
1	Petty Cash	Record cash top-ups	Log money added to the petty cash float with date, amount, source, and notes
2	Petty Cash	Record disbursements	Log money paid out with date, amount, purpose/expense category, and payee
3	Petty Cash	Running balance	Maintains and displays the current petty cash balance after every entry
4	Petty Cash	Single-currency amounts	All entries and vouchers are recorded in one currency, with no conversion needed
5	Petty Cash	Entry history	Full chronological list of all petty cash transactions, searchable and filterable
6	Journal Vouchers	Create JV	Create a journal voucher with JV number, date, notes, and supporting details
7	Journal Vouchers	Debit/credit lines	Record one or more debit and credit line items per voucher
8	Journal Vouchers	Approval workflow	JVs go through a review and approval step before being finalized
9	Journal Vouchers	Status tracking	Each JV carries a visible approval status (e.g., draft, pending, approved)
10	Journal Vouchers	JV history	Browse and search all past journal vouchers with their approval outcomes
11	Reports	Period summaries	Summarize cash movements and JV activity over a selected date range

12	Reports	Export statements	Download/export reports for reconciliation, bookkeeping, and audit purposes
13	Security	User authentication	Only registered, logged-in users can record or approve entries
14	General	Responsive UI	Works on desktop, tablet, and mobile devices
15	General	Data persistence	All entries and vouchers are stored securely in the cloud database

VIII. Screenshots of the system

Figure 1 - Branch Staff Dashboard

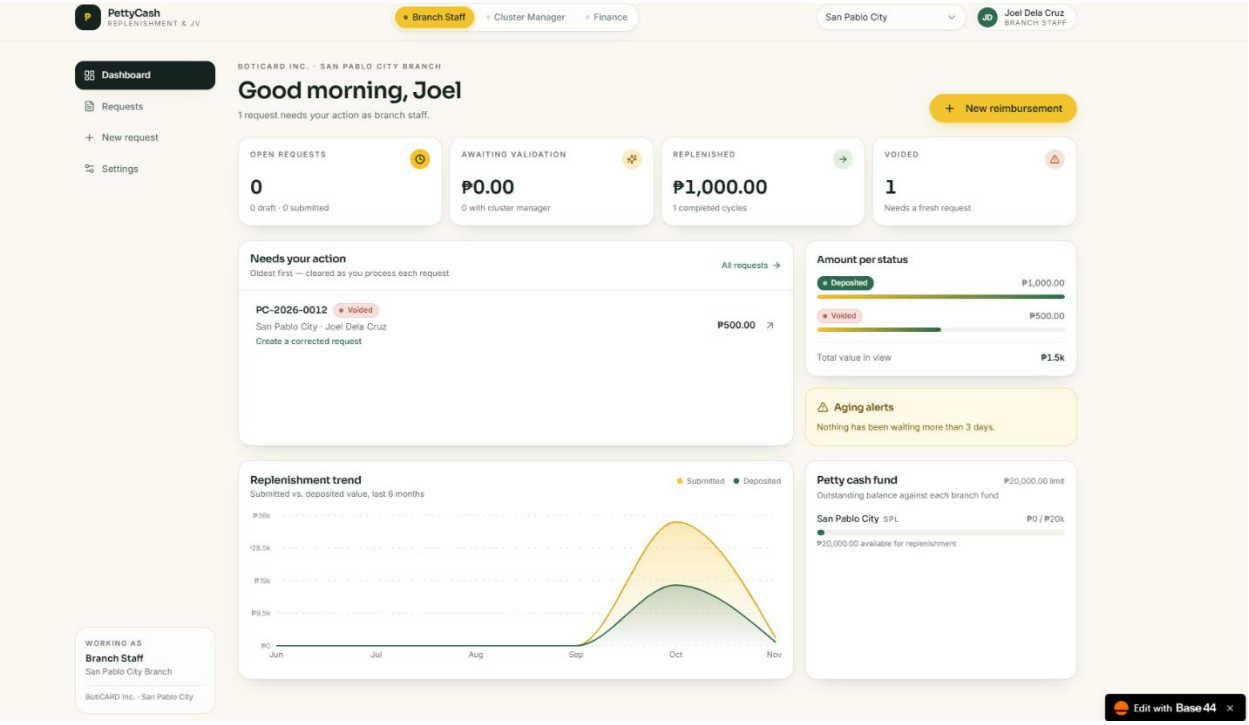


Figure 2 – Cluster Manager Dashboard

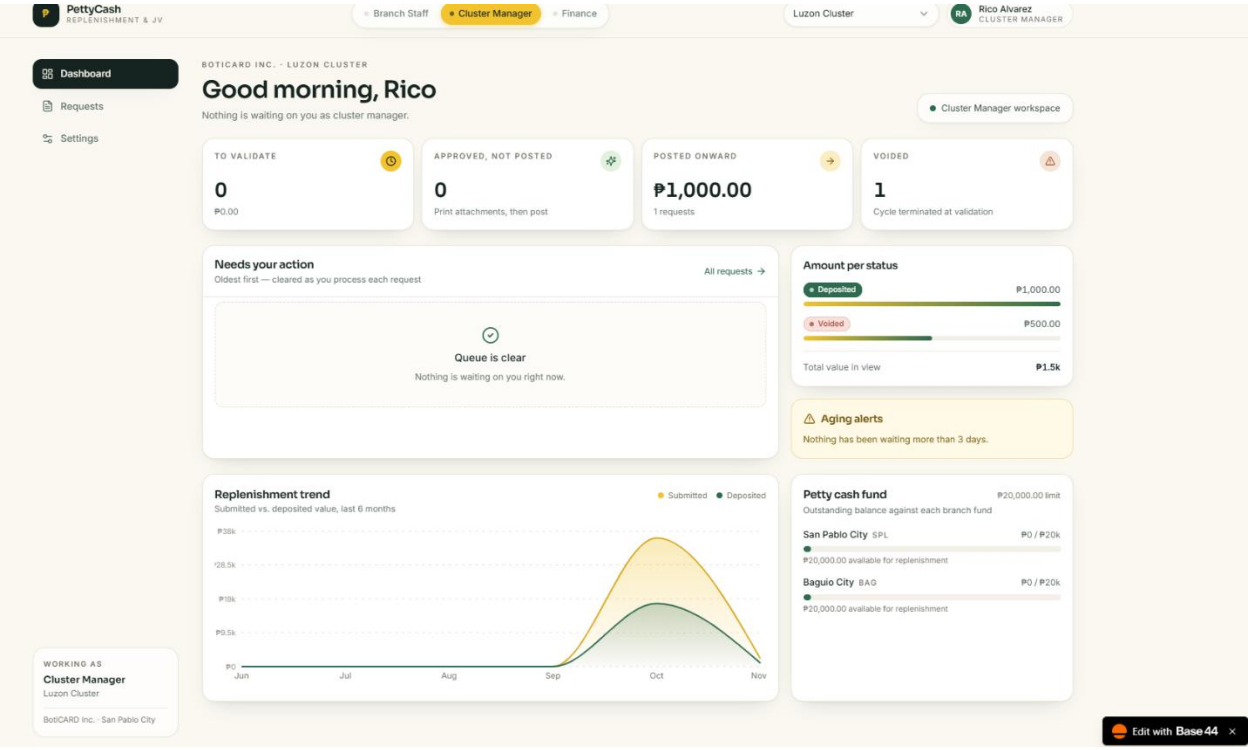


Figure 3 – Finance Dashboard

